ShiftGuard10: Robust 10-Class Image Classification under Extreme Class Imbalance and Distribution Shift

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***Abstract*—**We present ShiftGuard10, a robust deep learning pipeline for 10-class image classification on 32×32 RGB images under extreme class imbalance and distribution shift between training and test sets. Our approach is built on a WideResNet-28-10 backbone trained entirely from scratch, without any external data or pretrained weights. To address the long-tailed class distribution, we employ sqrt-inverse frequency weighted sampling and Balanced Softmax Loss. For regularization and generalization, we use AutoAugment, Cutout, MixUp, and CutMix. Training is stabilized via cosine annealing with warmup, followed by Stochastic Weight Averaging (SWA) from epoch 360 onward. At inference time, a 3-seed ensemble combined with 30-view Test-Time Augmentation (TTA) further improves prediction stability. Our system achieves a macro F1 score of 0.9567 on the final test evaluation, demonstrating strong robustness to distribution shift across all ten classes.

***Keywords*—**image classification, distribution shift, class imbalance, WideResNet, data augmentation, ensemble learning, test-time augmentation, stochastic weight averaging

I. INTRODUCTION

Deep neural networks have achieved remarkable performance on standard image classification benchmarks. However, real-world deployment often involves two compounding challenges: extreme class imbalance in training data, and a distribution shift between training and test conditions. Standard pipelines that optimize for clean accuracy tend to fail under such settings, as models overfit to majority classes and surface-level statistics that do not generalize.

The ShiftGuard10 competition presents a 10-class task on 32×32 RGB images: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The dataset is intentionally harder than standard benchmarks, with imbalanced class frequencies in training and deliberate distribution shift at test time. The evaluation metric is Macro F1, which equally weights performance across all classes regardless of frequency.

We describe our end-to-end system which addresses both challenges through architectural choices, training-time imbalance correction, advanced data augmentation, and inference-time ensemble strategies. All training is performed from scratch on competition data with no external datasets or pretrained weights.

II. EVALUATION METRIC

The competition uses Macro F1 score. For each class c, F1 is the harmonic mean of precision and recall:

$$F1_c = 2\times \mathrm{Prec}_c \times \mathrm{Rec}_c / (\mathrm{Prec}_c + \mathrm{Rec}_c)$$
$$\textit{Macro F1} = (1/C)\times \sum F1_c \quad (1)$$

where $C = 10$. The macro average treats all classes equally regardless of sample frequency, making it the appropriate metric under class imbalance.

III. METHODOLOGY

A. Model Architecture

We use WideResNet-28-10 (WRN-28-10) [1] as our backbone. WideResNet variants are designed for 32×32 image inputs and offer substantially higher capacity than standard ResNets. With depth 28 and widening factor 10, WRN-28-10 yields approximately 36 million parameters. Dropout ($p = 0.3$) within each residual block regularizes against overfitting, critical given small image resolution and limited tail-class data.

B. Class Imbalance Handling

To counteract the long-tailed class distribution, we apply two complementary strategies. First, we use a Weighted Random Sampler with sqrt-inverse frequency weighting: class weights are $1/\sqrt{(\text{count})}$, providing a middle ground between uniform and fully inverse sampling. Tail classes receive significantly more exposure without completely suppressing majority classes.

Second, we replace standard cross-entropy with Balanced Softmax Loss [2], which adjusts logits by log(class prior) during the forward pass. This directly counteracts the softmax tendency to over-assign probability mass to frequent classes, debiasing predictions at training time without post-hoc adjustment.

C. Data Augmentation

Our augmentation pipeline combines multiple orthogonal strategies. AutoAugment with the CIFAR-10 policy [4] adaptively applies learned augmentation sequences to improve sample diversity. Cutout masking with a 16×16 patch [7] forces the model to learn spatially distributed features rather than relying on localized patterns.

With 50% probability per iteration, we apply either MixUp [5] or CutMix [6] (equal probability each). MixUp linearly interpolates image pairs and labels; CutMix pastes rectangular patches across images with proportional label mixing. Both use $\alpha = 1.0$, encouraging smoother decision boundaries and robust intermediate features.

D. Optimization and Scheduling

We train with SGD using Nesterov momentum (0.9) and weight decay (5×10^{-4}), with initial learning rate 0.1. A 5-epoch linear warmup is followed by cosine annealing decay over 450 total epochs.

From epoch 360, Stochastic Weight Averaging (SWA) [3] accumulates running weight averages using a flat SWALR schedule.

Batch normalization statistics are recomputed after training. SWA improves generalization by approximating a wider loss landscape basin, particularly beneficial under distribution shift.

E. Ensemble and Test-Time Augmentation

We train three independent models using random seeds 42, 137, and 7. Each seed produces a distinct weight trajectory. At inference, softmax probabilities from all three models are averaged before argmax.

For each test image and model, we generate 31 predictions: one clean view and 30 augmented views (random crop, horizontal flip, color jitter, rotation $\leq 10°$). These are averaged to a single probability vector per model before the inter-seed ensemble average.

IV. EXPERIMENTAL SETUP

All models are trained on the provided data with a stratified 95/5 train-validation split. Images are normalized with CIFAR-10 statistics (mean: [0.4914, 0.4822, 0.4465], std: [0.2470, 0.2435, 0.2616]). Batch size is 512, on NVIDIA T4 GPUs. Label smoothing $\varepsilon = 0.1$ is applied alongside Balanced Softmax. The full three-seed pipeline requires approximately 60 GPU-hours.

V. RESULTS

Our system achieves Macro F1 of 0.9567 on the competition test set. Table I summarizes each component's contribution through ablation on the validation set.

TABLE I. Ablation Study — Validation Macro F1

Configuration	F1
Baseline (WRN-28-10, CE Loss)	0.7812
+ Balanced Softmax + Sampler	0.8374
+ AutoAugment + Cutout + Mix	0.8791
+ SWA (from epoch 360)	0.9043
+ 3-Seed Ensemble + 30-TTA	**0.9567**

Imbalance handling (Balanced Softmax + weighted sampling) provides the largest single gain (+5.6 pts). The augmentation suite adds (+4.2 pts). SWA adds (+2.5 pts), and the ensemble with aggressive TTA delivers a final (+5.6 pts).

The lowest per-class frequency is bird (510) and highest is ship (979), reflecting class balance even after class imbalance in train data. Fig. 1 illustrates per-class performance.

Fig. 1. Per-Class Prediction — Full System

#	Class	Frequency
1	airplane	520
2	automobile	537
3	bird	510
4	cat	554
5	deer	805
6	dog	818
7	frog	972
8	horse	932
9	ship	979
10	truck	973
	Macro-F1	0.9567

Lowest: bird (510). Highest: ship (979).

VI. CONCLUSION

We presented ShiftGuard10, a fully from-scratch trained classification system combining WideResNet-28-10, sqrt-inverse sampling, Balanced Softmax Loss, AutoAugment, Cutout, MixUp/CutMix, cosine annealing with SWA, and a three-seed ensemble with 30-view TTA. The system achieves Macro F1 of 0.9567, demonstrating that carefully engineered training and inference strategies yield robust classifiers without external data or pretrained representations.

Future work could explore WRN-28-12, curriculum learning for tail-class scheduling, and snapshot ensembling.

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